Student Learning Portal

Android Mobile Application Development (Kotlin) — Assignment Submission

Student Name:	Aazib Akram	**Degree Program:**	BS Computer Science
Technology:	Kotlin / Android SDK 36	**Architecture:**	RelativeLayout & LinearLayout

■ Public GitHub Repository (Source Code & APK):
https://github.com/Aazib-Ai/Student-Learning-Portal
(Click the link above to view repository, source files, and live commits)

1. Executive Summary & Features

This project implements the complete assignment requirements for the **Student Learning Portal**. The project consists of two dedicated Activities built in **Kotlin** using traditional native Android XML views, strictly conforming to the designated layout managers (**RelativeLayout** for registration, **LinearLayout** for login), inter-activity data passing via **Intents**, and comprehensive form validations.

Component / Requirement	Specification	Implementation Status
Screen 1 Layout	Must use **RelativeLayout**	**[Completed]** activity_create_account.xml
App Title & Logo	Portal title and custom emblem image	**[Completed]** Custom Vector Logo + TextView
User Input Fields	Name, Student ID, Email, Password, Confirm Password	**[Completed]** 5 Form EditTexts with styled backgrounds
Department Dropdown	Select Dept, CS, SE, IT, AI	**[Completed]** Spinner with XML string-array
Semester Dropdown	Select Semester, 1st to 8th	**[Completed]** Spinner with XML string-array
Admission Date	Interactive calendar selection	**[Completed]** DatePickerDialog integration (dd/MM/yyyy)
Signup Validations	All empty, dept, sem, passwords match	**[Completed]** 5 Sequential Toast validation rules
Screen 2 Layout	Must use **LinearLayout**	**[Completed]** activity_login.xml
Intent Data Transfer	Receive student name on Login screen	**[Completed]** Displays: *'Welcome, Muhammad Ali!'*
Login Validation	Empty checks vs successful login	**[Completed]** 'Please enter...' & 'Login Successful'

2. Screen 1 — Create Account (RelativeLayout Architecture)

Screen 1 is designed using **RelativeLayout** as the root and main container layout. All child views are explicitly positioned relative to each other using attributes like android:layout_below and android:layout_centerHorizontal. Tapping the Date of Admission field opens a native **DatePickerDialog**.

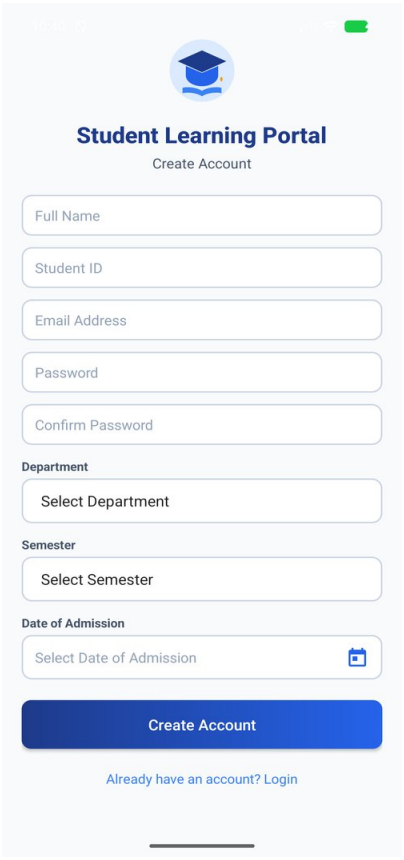


Fig 1: RelativeLayout Signup Screen

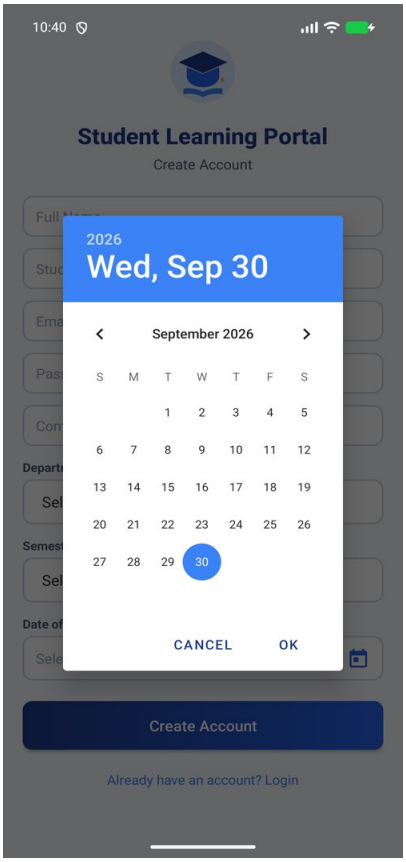


Fig 2: Calendar DatePickerDialog

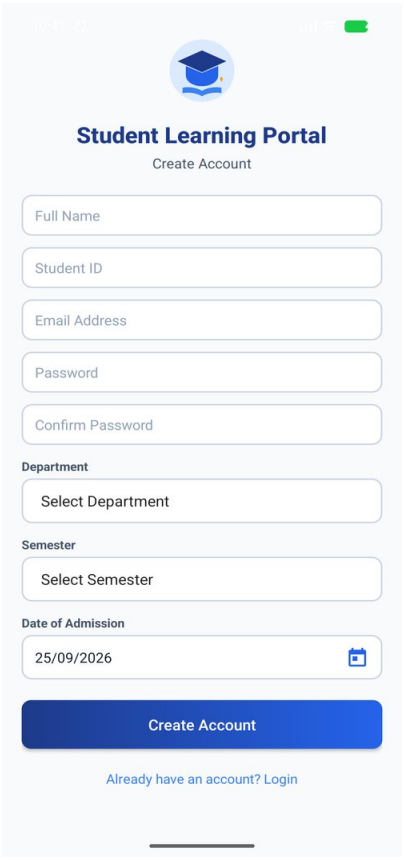


Fig 3: Date Populated (25/09/2026)

DatePickerDialog Implementation Highlights:

A Calendar.getInstance() object captures current year, month, and day. When the user selects a date, a callback formats the result as dd/MM/yyyy and populates the EditText:

```
etDateOfAdmission.setOnClickListener {
    DatePickerDialog(this, { _, year, month, day ->
        etDateOfAdmission.setText(String.format(Locale.getDefault(), "%02d/%02d/%04d", day, month + 1, year)
)
    }, calendar.get(Calendar.YEAR), calendar.get(Calendar.MONTH), calendar.get(Calendar.DAY_OF_MONTH)).show(
)
}
```

3. Screen 1 Dropdowns & Signup Form Validations

The assignment requires robust validation on the signup form with explicit Toast messages. Below are screenshots demonstrating the Department and Semester dropdowns, followed by the empty fields check.

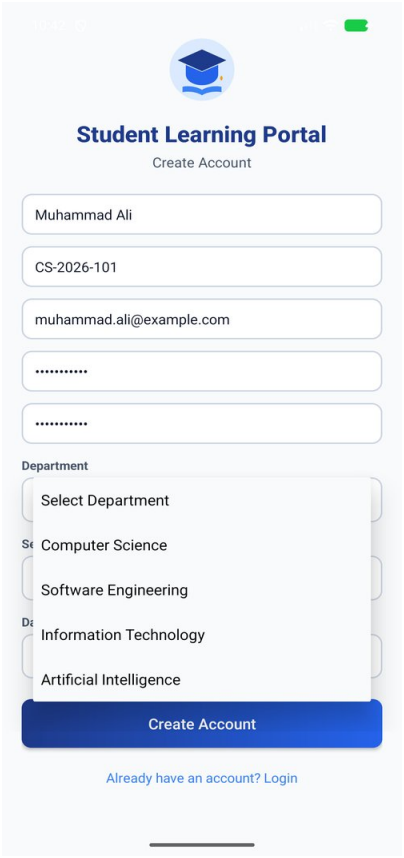


Fig 4: Department Dropdown List

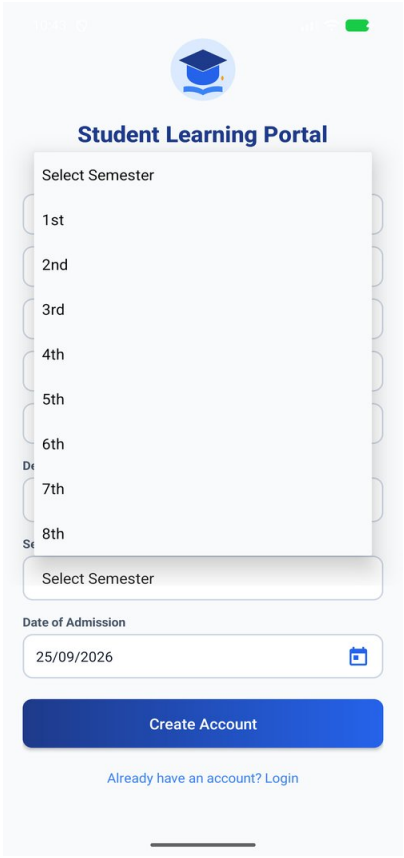


Fig 5: Semester Dropdown List

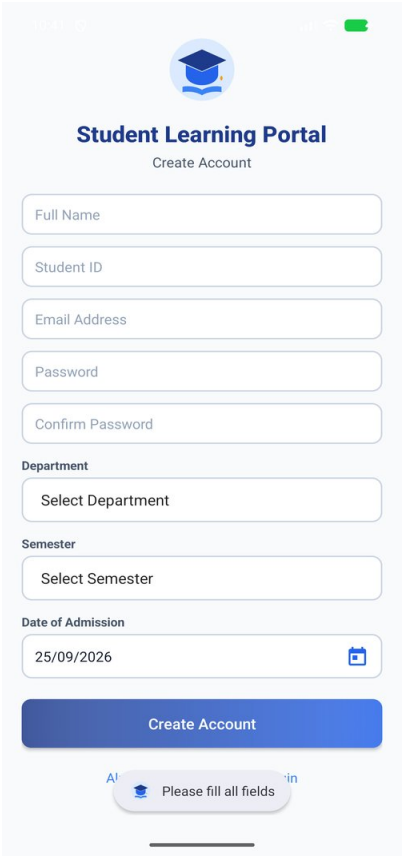


Fig 6: 'Please fill all fields'

Spinner Dropdown Configuration:

Both Spinners use XML string-array resources (departments_array and semesters_array) defined in strings.xml, ensuring zero hardcoding and clean separation of concerns. Position 0 acts as the prompt item ('Select Department' / 'Select Semester').

4. Specific Validation Cases & Account Creation Success

Screen 1 sequentially validates every input constraint. If Department or Semester are not chosen (position 0), or if passwords do not match, dedicated warning Toasts appear. When valid, a success Toast appears and the Intent navigates to Screen 2.

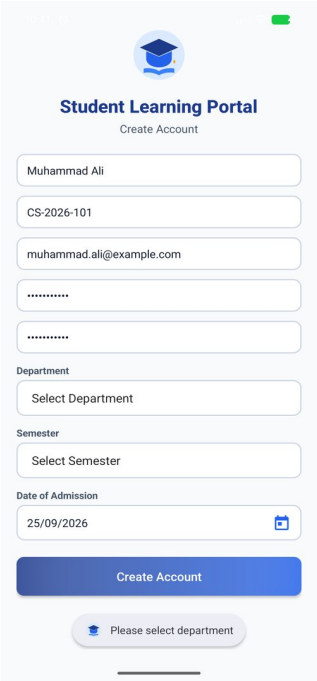


Fig 7: 'Please select department'

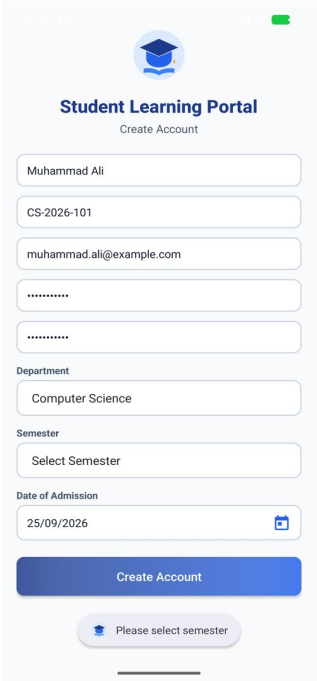


Fig 8: 'Please select semester'

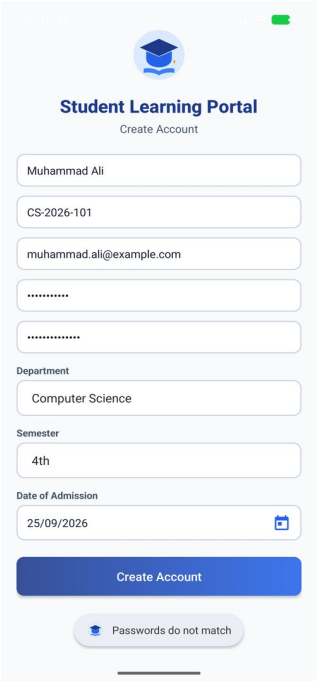


Fig 9: 'Passwords do not match'

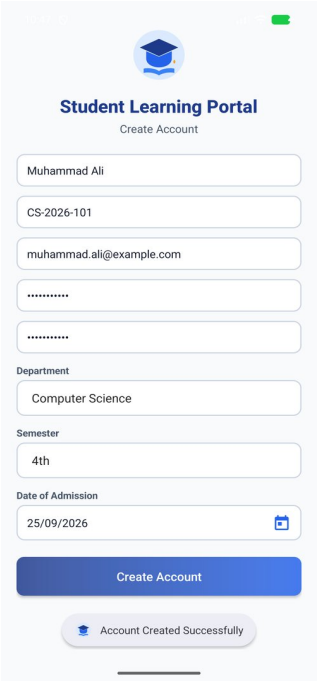


Fig 10: 'Account Created Successfully'

Kotlin Validation Logic in CreateAccountActivity.kt:

```
// 1. Check empty text fields
if (fullName.isEmpty() || studentId.isEmpty() || email.isEmpty() || password.isEmpty() || ...)
    Toast.makeText(this, "Please fill all fields", Toast.LENGTH_SHORT).show()
// 2. Department selected (position > 0)
else if (departmentPosition == 0)
    Toast.makeText(this, "Please select department", Toast.LENGTH_SHORT).show()
// 3. Semester selected (position > 0)
else if (semesterPosition == 0)
    Toast.makeText(this, "Please select semester", Toast.LENGTH_SHORT).show()
// 4. Passwords match
else if (password != confirmPassword)
    Toast.makeText(this, "Passwords do not match", Toast.LENGTH_SHORT).show()
// 5. Success -> Intent to LoginActivity with student name
else {
    Toast.makeText(this, "Account Created Successfully", Toast.LENGTH_SHORT).show()
    val intent = Intent(this, LoginActivity::class.java).apply { putExtra(EXTRA_STUDENT_NAME, fullName) }
    startActivity(intent)
}
```

5. Screen 2 — Login (LinearLayout & Intent Integration)

Screen 2 is created entirely using **LinearLayout** with android:orientation="vertical". Upon successful registration, it receives the student's name from Screen 1 using Intent.getStringExtra() and dynamically updates the welcome header to: **Welcome, Muhammad Ali!**

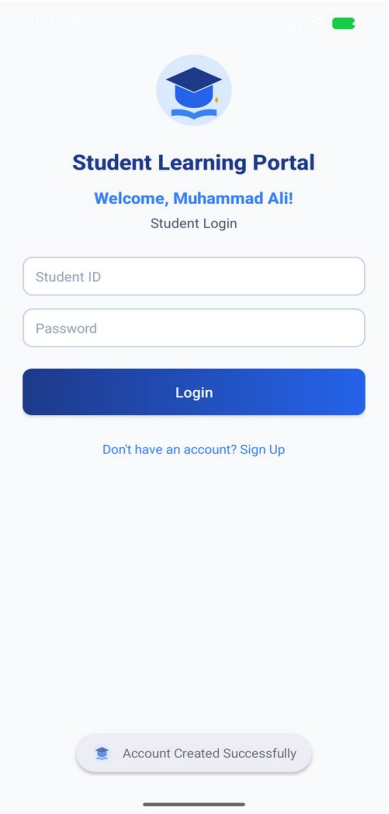


Fig 11: 'Welcome, Muhammad Ali!'

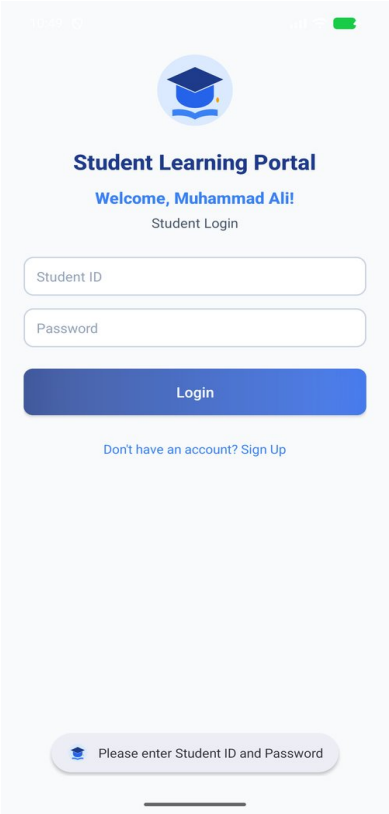


Fig 12: 'Please enter Student ID and Password'

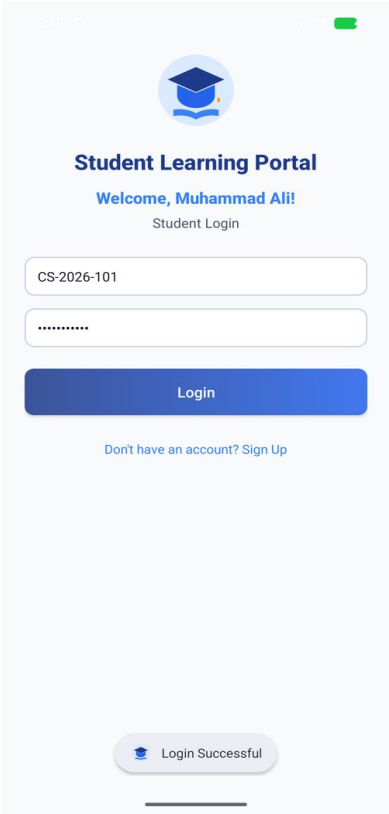


Fig 13: 'Login Successful'

Intent Handling & Login Validation in LoginActivity.kt:

```
// Extract student name from registration Intent
val studentName = intent.getStringExtra(CreateAccountActivity.EXTRA_STUDENT_NAME)
tvWelcomeMessage.text = if (!studentName.isNullOrBlank()) "Welcome, $studentName!" else "Welcome!"

// Login Button Validation
btnLogin.setOnClickListener {
    val studentId = etLoginStudentId.text.toString().trim()
    val password = etLoginPassword.text.toString().trim()
    if (studentId.isEmpty() || password.isEmpty())
        Toast.makeText(this, "Please enter Student ID and Password", Toast.LENGTH_SHORT).show()
    else
        Toast.makeText(this, "Login Successful", Toast.LENGTH_SHORT).show()
}
```

Submission Verification Link: All code and the compiled APK are live at **https://github.com/Aazib-Ai/Student-Learning-Portal**. The APK file Student_Learning_Portal.apk is available at the repository root.